Homework 5

Calculus I (STM 1001)

due: **Friday, March 13, 2026 at 11:30a**

Instructions: Please submit solutions on Populi as a knitted markdown pdf. For handwritten work, please scan your solution pages and include in the final pdf. Homeworks can be completed individually or in groups of up to three people. Names of all group members must be included on the writeups.

Part I: Course Review

The following problems review key concepts from the entire course. They are designed to help you synthesize what we've learned from functions and derivatives through optimization and gradient descent.

Problem 1: Functions, parameters, and fitting

a. Pattern-book functions. The function $p(x) = \dfrac{1}{1+e^{-(x-5)/2}}$ is a shifted and scaled sigmoid.

- What is the horizontal asymptote as $x \to \infty$? As $x \to -\infty$?
- Evaluate $p(5)$. What is special about this point?
- In `R`, plot $p(x)$ over $[-2, 12]$. Which pattern-book function is this, and what do the parameters (the 5 and the 2) control?

b. Assembling functions. A model for daily energy demand (in gigawatts) is

$$E(t) = 8 + 3\sin\left(\frac{2\pi(t-6)}{24}\right) + 2e^{-(t-14)^2/8},$$

where t is hours after midnight ($0 \le t \le 24$).

- Identify the three "building-block" pattern-book functions that are combined to form $E(t)$. (Constant, sinusoid, hump, etc.)
- What does the term $3\sin(\cdots)$ represent? What does the term $2e^{-(t-14)^2/8}$ represent?
- In `R`, plot $E(t)$ over $[0, 24]$. At approximately what time(s) does demand peak?

c. Fitting a parameter. A cup of tea cools according to $T(t) = 22 + (T_0 - 22)e^{-kt}$, where T_0 is initial temp and $k > 0$ is the cooling rate. You measure $T(0) = 85$ and $T(15) = 58$ (room temp is 22°C).

- Solve for k using the two measurements.
- Predict $T(30)$. At what time will the tea reach 35°C?

Problem 2: Derivatives, concavity, and optimization

a. Let $f(x) = x^4 - 4x^3 + 6$.

- Find $f'(x)$ and $f''(x)$.

- Find all critical points. Use the second derivative test to classify each as local max, local min, or neither.
- Find all inflection points. On which intervals is f concave up? Concave down?
- In `R`, plot $f(x)$ over a domain that shows all key features. Annotate the critical and inflection points.

b. Position, velocity, acceleration. A drone's height (in meters) is $h(t) = 20t - 5t^2$ for $t \geq 0$.

- Find $v(t) = h'(t)$ and $a(t) = h''(t)$. When does the drone reach its maximum height? What is that height?
- At $t = 1$, is the drone going up or down? Is it speeding up or slowing down? Justify using $v(1)$ and $a(1)$.

c. Tangent line approximation. Let $g(x) = \ln(x)$.

- Find the equation of the tangent line to g at $x = 1$.
- Use your tangent line to approximate $\ln(1.1)$. Compare to the true value (use a calculator or `R`).

Problem 3: Multivariable calculus and optimization

a. Let $f(x, y) = x^2 + 2xy + 2y^2 - 4x - 6y + 10$.

- Find $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$. Write the gradient ∇f.
- Find all critical points by setting $\nabla f = \mathbf{0}$.
- Compute the Hessian and use the second derivative test to classify the critical point(s).

b. Let $h(x, y) = e^{-(x^2+y^2)}$.

- Find $\dfrac{\partial h}{\partial x}$ and $\dfrac{\partial h}{\partial y}$. (Use the chain rule.)
- Evaluate ∇h at $(1, 0)$ and at $(0, 0)$. In which direction does h increase fastest at each point?

c. Gradient descent. For $f(x, y) = x^2 + y^2$, perform one step of gradient descent starting at $(2, 1)$ with learning rate $\alpha = 0.3$.

- Compute $\nabla f(2, 1)$ and the new point (x_1, y_1).
- Verify that $f(x_1, y_1) < f(2, 1)$.

Part II: Neural Networks and the Chain Rule

These problems connect the calculus we've learned to how neural networks are trained.

Problem 4: Chain rule through a composition

Consider the composite function $L = \frac{1}{2}(u-5)^2$ where $u = \sigma(z)$ and $z = 2w+3$, with $\sigma(z) = \frac{1}{1+e^{-z}}$ (the sigmoid).

a. Draw the "computational graph": $w \to z \to u \to L$. Label each arrow with the operation (e.g., "$\times 2$, $+3$", "σ", etc.).

b. Compute $\dfrac{\partial L}{\partial w}$ using the chain rule in three steps:

- $\dfrac{\partial L}{\partial u} = ?$
- $\dfrac{\partial u}{\partial z} = ?$ (Recall: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.)
- $\dfrac{\partial z}{\partial w} = ?$

c. Combine: $\dfrac{\partial L}{\partial w} = \dfrac{\partial L}{\partial u} \cdot \dfrac{\partial u}{\partial z} \cdot \dfrac{\partial z}{\partial w}$.

d. For $w = 0$, compute z, u, and $\dfrac{\partial L}{\partial w}$ numerically. Show each step.

Problem 5: Single neuron with ReLU activation

The ReLU (Rectified Linear Unit) activation is $\phi(z) = \max(0, z)$. For $z > 0$, $\phi'(z) = 1$; for $z < 0$, $\phi'(z) = 0$; at $z = 0$ the derivative is undefined (we take $\phi'(0) = 0$ by convention).

A single neuron with one input uses $\hat{y} = \phi(wx + b)$. For one data point (x, y), the MSE loss is $L = \frac{1}{2}(\hat{y} - y)^2$.

a. Chain rule for ReLU. Let $z = wx + b$. Using the chain rule, show that when $z > 0$:

$$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot x, \qquad \frac{\partial L}{\partial b} = (\hat{y} - y).$$

(When $z \leq 0$, $\hat{y} = 0$ and the gradients are zero—this is known as the "dead neuron" problem.)

b. Numerical example. For $(x, y) = (2, 3)$ and $(w, b) = (1, -1)$:

- Compute z, $\hat{y}$, and L.
- Compute $\dfrac{\partial L}{\partial w}$ and $\dfrac{\partial L}{\partial b}$.
- Perform one gradient descent step with $\alpha = 0.2$. Report the new (w, b) and new $\hat{y}$. Did $\hat{y}$ move toward $y = 3$?

c. In `R`: Implement gradient descent for this ReLU neuron. Use data $(x, y) = (2, 3)$, initial $(w, b) = (1, -1)$, $\alpha = 0.2$, and run 15 iterations.

- Report the final (w, b) and final $\hat{y}$.

- Plot L versus iteration. Does it decrease? (If z ever becomes ≤ 0, the neuron "dies" and gradients become zero—note if that happens.)

Problem 6: Backpropagation — two-layer network

Consider a tiny network: one input x, one hidden neuron with sigmoid σ, one output neuron (linear, no activation). The forward pass is:

$$z_1 = w_1 x + b_1, \quad a_1 = \sigma(z_1), \quad \hat{y} = w_2 a_1 + b_2.$$

For one data point (x, y), the loss is $L = \frac{1}{2}(\hat{y} - y)^2$.

a. Forward pass. For $x = 1$, $y = 0.8$, $w_1 = 0.5$, $b_1 = 0$, $w_2 = 1$, $b_2 = 0$:

- Compute z_1, a_1, and $\hat{y}$.
- Compute L.

b. Backward pass — output layer. The gradient of L with respect to $\hat{y}$ is $\dfrac{\partial L}{\partial \hat{y}} = \hat{y} - y$.

- Show that $\dfrac{\partial L}{\partial w_2} = (\hat{y} - y) \cdot a_1$ and $\dfrac{\partial L}{\partial b_2} = \hat{y} - y$.
- Compute $\dfrac{\partial L}{\partial a_1}$ (the "error signal" passed to the hidden layer). (Hint: L depends on $\hat{y}$, and $\hat{y} = w_2 a_1 + b_2$, so $\dfrac{\partial L}{\partial a_1} = \dfrac{\partial L}{\partial \hat{y}} \cdot w_2$.)

c. Backward pass — hidden layer. Using the chain rule (as in the single-neuron case), show that

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial a_1} \cdot a_1(1 - a_1) \cdot x, \qquad \frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial a_1} \cdot a_1(1 - a_1).$$

d. Why "back" propagation? In one or two sentences, explain why we compute gradients from the output layer backward to the input layer, rather than the other way around.

Problem 7: Synthesis — gradient descent in `R`

In `R`, implement gradient descent to minimize $f(x, y) = (x - 2)^2 + 2(y - 1)^2$.

a. By hand, find the minimum of f (set $\nabla f = \mathbf{0}$ and solve). What is the minimizer (x^*, y^*) and the minimum value $f(x^*, y^*)$?

b. Write a loop that starts at $(0, 0)$, uses learning rate $\alpha = 0.3$, and runs 20 iterations. Report the final (x, y) and $f(x, y)$. How close are you to the true minimum?

c. Plot the path of (x, y) over the 20 iterations on a contour plot of f. (Use `contour_plot()` or `ggplot2` with `stat_contour`.) Does the path move toward $(2, 1)$?

d. Optional: Try $\alpha = 0.1$ and $\alpha = 0.8$. What happens? (Too small: slow convergence; too large: overshooting.)